



Department of Mathematics  
The Mathematics Community (TMC)

 Introduction to Version Control Systems  
Git, GitHub & Collaboration

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# Outline

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- 2 Git & GitHub
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# What is Version Control?

## Definition

A **Version Control System (VCS)** tracks and manages changes to files over time, maintaining a detailed history of every modification.

### ☰ Centralized VCS (CVCS)

- Single central server stores all files and history
- Developers commit/pull directly to the server
- Examples: *Apache Subversion (SVN)*, *CVS*

### ☐☐ Distributed VCS (DVCS)

- Every collaborator holds a **full copy** of the project including its history
- Works offline; more resilient
- Examples: *Git*, *Mercurial*, *Darcs*





## Git

- Free, open-source DVCS
- Created by **Linus Torvalds** in 2005
- Handles projects of any size with speed
- Allows simultaneous multi-developer work without overwriting changes
- Runs **locally** on your machine



## GitHub

- Online platform for hosting Git repositories in the **cloud**
- Enables collaboration, code review, and issue tracking
- Git and GitHub are complementary tools

*“Git tracks your history locally — GitHub shares it with the world.”*



# Installing Git — Windows

## Step 1: Download

- Visit [git-scm.com](https://git-scm.com) and download the **64-bit Windows Setup .exe**

## Step 2: Install

- Run the installer, accept the license, and click *Next*

## Step 3: Configure PATH

- Select *"Git from the command line and also from 3rd-party software"*

## Step 4: Verify

```
git --version
```

## Step 5: Set credentials

```
git config --global user.name "Your Name"  
git config --global user.email "you@example.com"
```

# Installing Git — Linux

## Open a terminal and run

```
sudo apt update && sudo apt upgrade  
sudo apt install git
```

## Confirm the installation

```
git --version
```

## Configure your identity

```
git config --global user.name "Your Name"  
git config --global user.email "you@example.com"
```

**NB:** These credentials are attached to every commit you make, so set them correctly from the start.

# Initializing a Repository

Create a new Git repository in your project folder

Run the command below to initialize a new git repository.

```
git init
```

After initializing, explore with these commands:

| Command    | Description                                       |
|------------|---------------------------------------------------|
| git status | Current state of working directory & staging area |
| git log    | Chronological list of all commits                 |
| git --help | Display help for Git commands                     |



# The Core Workflow

## Staging & Committing

`git add .` Moves all changes to the **staging area** (index) — a draft board between your working directory and permanent history.

`git commit -m "..."` Saves a **permanent snapshot** of staged changes. Use a meaningful message to describe what changed.

```
git add .  
git commit -m "Describe what changed"  
git push
```

## Best Practice

Commit often with clear messages. You can always restore your project to any snapshot.



# Method 1: GitHub CLI (gh)

## Install on Windows

```
winget install GitHub.cli    # using winget  
scoop install gh            # using Scoop
```

## Authenticate

```
gh auth login
```

## Install on Windows

## Create & push a new repo in one command

```
gh repo create YOUR_PROJECT_NAME \  
  --public --source=. --remote=origin --push
```

**NB:** Replace "--public" with "--private" if you do not want the repository visible to

## Method 2: Clone an Existing Repository

- 1 Log in to github.com
- 2 Click **New** on the Repositories tab
- 3 Fill in name, description, and visibility, then create
- 4 Clone the repository locally:

```
git clone https://github.com/username/repository.git
```

Add your files, then push:

```
git add .  
git commit -m "Initial Commit"  
git push --set-upstream origin master
```



## Method 3: Remote Add (Existing Local Project)

Already have a local project? Link it to a new empty GitHub repo:

```
git init
git remote add origin YOUR_REPOSITORY_URL
git add .
git commit -m "Initial Commit"
git push --set-upstream origin master
```

### Tip

YOUR\_REPOSITORY\_URL looks like:

<https://github.com/username/repository.git>

### Name Match

Your local project folder and GitHub repository should share the same name for clarity.

# Summary

## Key Concepts

- VCS = change tracking over time
- DVCS (Git) vs CVCS (SVN)
- Git runs locally; GitHub hosts in the cloud

## Essential Commands

- `git init` — start a repo
- `git add .` — stage changes
- `git commit -m` — save snapshot
- `git push` — upload to GitHub

## 3 Ways to Connect to GitHub

- 1 **gh CLI** — fastest for new projects
- 2 **Clone** — start from GitHub
- 3 **Remote add** — link an existing local project

## Next Steps

Explore README.md files, .gitignore, branching (`git branch`), and pull requests on GitHub.





# Thank You!

Start tracking your code today.



[git-scm.com](https://git-scm.com)



[github.com](https://github.com)



`git --help`

